



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.

--	--	--	--	--	--

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY(COMPUTER SCIENCE AND ENGINEERING)

Batch: 2021-25,2022-26,2023-27

Semester: III

Course: Digital Electronics and Logic Design

Course Code: 0705210307

Date: 06/12/2024

Day: Friday

Maximum Marks: 45

Time: 13:30 - 15:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

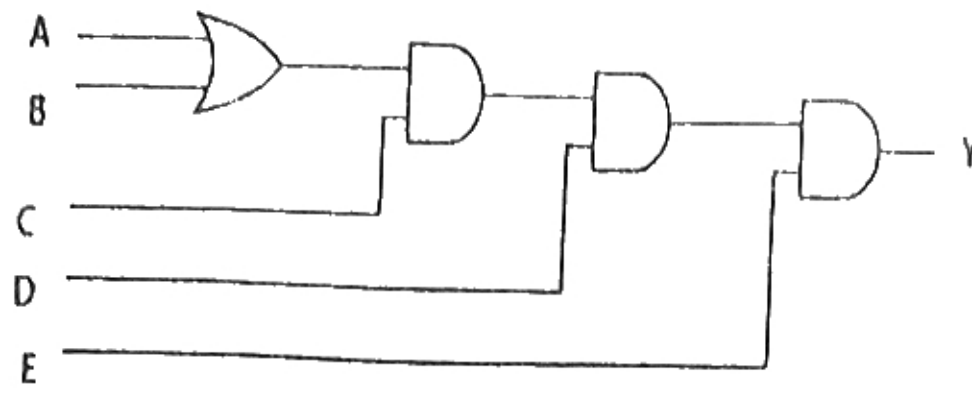
- Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
- Write your seat number in words and figures in the Answer sheet provided to you.
- Do not copy or help anyone in person or by using signs/ digital gadgets while you write the exam/ while you start the exam/ while you end the exam.
- Calculator is allowed.
- Figures to the rights indicate maximum marks.
- This question paper comprises of 2 printed sides.

- Q1
- Convert $(EB7)_{16}$ into binary and octal number system. 3 CO1 BL3
 - Perform $(67)_{10} - (34)_{10}$ in binary using 2's complement method. 3 CO1 BL3
 - Represent the following numbers in its 1's complement form.
+6 and -6 2 CO1 BL2
 - Find the hexadecimal addition of 23 and 4A. 2 CO1 BL3
- Q.2
- Design AND gate using NAND and NOR gate. 4 CO2 BL6
 - State and explain De-Morgan's theorem. 3 CO2 BL2

OR

Derive the logical expression for the following circuit and simplify.

3 CO2 BL4



Q.3 a) Convert the following equations in to its Canonical SOP form
 $Y = (ABC + AB + AC)$

2 CO3 BL3

b) Minimize the following equation using K-Map & Implement the optimized circuits using gates.

4 CO3 BL3

$$F(A, B, C, D) = \sum M(6, 7, 8, 9) + d(12, 13, 14, 15)$$

c) Differentiate Weighted and Non-Weighted Code.

2 CO3 BL4

Q.4 a) Illustrate the pin diagram of 3:8 Decoder IC74138.

3 CO4 BL2

b) Design a Full Adder circuit with its truth table.

3 CO4 BL6

OR

Explain magnitude comparator IC 7485 in details.

3 CO4 BL2

c) Define De-multiplexer and implement 1:16 DEMUX using 1:4 DEMUX.

4 CO4 BL4

Q.5 a) Draw and explain clocked T flip flop.

3 CO5 BL2

b) Discuss synchronous ring counter in detail.

4 CO5 BL2

c) Convert SR flip flop to JK flip flop.

3 CO5 BL3

OR

Differentiate buffer and shift register.

3 CO5 BL4



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under Section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A' Grade (3.58/4) Awarded Category - I by UGC

Seat No.					
----------	--	--	--	--	--

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2021-25

Semester: II

Course: Programming and Problem Solving

Course Code: 0705210206

Batch: 2021-25 (Sem II)

Backlog October/Odd 2022 Examination

Date: 09/01/2023

Maximum Marks: 30

Day: Monday

Time: 10:00 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

- Q.1 a) Illustrate different phases of Computational thinking with Example. 3 CO1, BL4
- b) Write an algorithm to find factorial of a number using recursion. 3 CO1, BL3
- Q.2 a) Predict the output and justify your answer: 3 CO2, BL3
- i) `-67%7`
 - ii) `6.6//6`
 - iii) `(90-70) * 10/5`
 - iv) `not False`
 - v) `type((4,5))`
 - vi) `5*1**2`
- b) Describe any two list operations and list methods. Write a Python program to check whether an element is present in the list or not using membership operator. 3 CO2, BL2
- Q.3 a) Write a Python program to prompt a number from user and find reverse of number using recursion 4 CO3, BL3
- b) Discuss about any two methods for passing parameters to a function with examples. 4 CO3, BL2

Q.4 a) Create a class named School having class attribute as SchoolCity. 5 CO4, BL5
Instance attributes as StudName and StudStd. The Student class
should have a method as SetAge(), that prompts age from user and
displays the age.

b) Create a child class that will inherit all the features of the School 5 CO4, BL5
class used in Q.4 (a) and have additional attribute as
SchoolBranch. Also, create show() method to display all the
features of School class.



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2022-26

Semester: I

Course: Physics

Course Code: 0705210102

Date: 20/12/2023

Maximum Marks: 45

Day: Wednesday

Time: 10:00 - 11:30

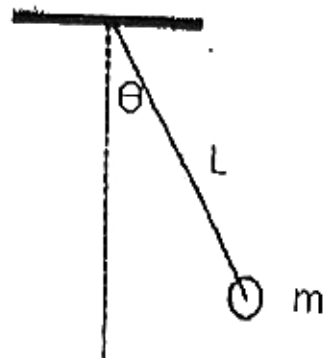
NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

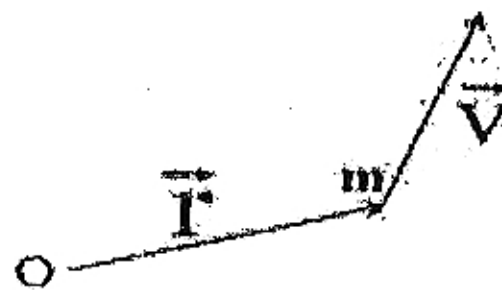
- Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
- Write your seat number in words and figures in the Answer sheet provided to you.
- Do not copy or help anyone in person or by using signs/ digital gadgets while you write the exam/ while you start the exam/ while you end the exam.
- Calculator is allowed.
- Figures to the rights indicate maximum marks.
- This question paper comprise of 2 printed sides.
- All questions are compulsory.

- Q1
- | | | |
|--|--------|---------|
| a) Define population inversion? | 1Marks | CO5 BL2 |
| b) Write any two medical applications of laser? | 2Marks | CO5 BL2 |
| c) A beam of an electron is deflected by magnetic field having direction perpendicular to the velocity of the electrons, what would be effect on their kinetic energy? | 2Marks | CO6 BL2 |
| d) Write the conditions of Bright fringes (maxima) and Dark fringes (minima) for reflected light from the transparent film having refractive index μ , thickness t and angle of refraction ϕ | 2Marks | CO5 BL3 |
| e) Newton's rings observed with wavelength 6.0×10^{-7} meter and diameter of the 10th dark ring is 0.5 cm. find the radius of curvature of the lens? | 3Marks | CO5 BL3 |
| f) A parallel beam of light incident on a plane transmission grating having 1000 lines/cm and first order spectral line is observed to be deviated by 30° calculate the wavelength of spectral lines? | 3Marks | CO6 BL4 |
- Q2
- | | | |
|---|--------|---------|
| a) Define zeroth law of thermodynamics and explain using block diagram? | 4Marks | CO4 BL1 |
| b) Explain the working of nuclear reactor and give names of its components? | 4Marks | CO5 BL3 |

- Q3 a) Derive the time period of the simple pendulum having a point mass m , suspended by a light string of length L . Point mass is pulled one side and released for oscillations. You can approximate $\sin \theta = \theta$ for your derivations. 4Marks CO1 BL3



- b) Define critical temperature T_c of the superconductor and plot a graph of electrical resistivity versus temperature T showing the transition of normal conductor to superconductor? 3Marks CO7 BL1
- c) Calculate the angular momentum $\vec{L}$ and torque $\vec{\tau}$ of a particle of mass $m = 5 \text{ kg}$ about the point O having velocity $\vec{v} = 2\hat{i} + 4\hat{j}$ as shown in the figure below. Where position vector of the particle from point O is $\vec{r} = 3\hat{i} + 5\hat{j}$. 4Marks CO3 BL4



- Q4 a) Explain magnetostriction effect ? 3Marks CO7 BL2
- b) Discuss the physical properties of nanomaterials? 3Marks CO7 BL2
- Q5 a) Explain Hall Effect using detailed diagram? 3Marks CO7 BL2
- b) Classify the materials (conductor, semiconductor, and insulators) on the basis of band diagram? 4Marks CO7 BL2



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.

--	--	--	--	--	--

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2022-26

Semester: I

Course: Physics

Course Code: 0705210102

Date: 23/05/2024

Maximum Marks: 45

Day: Thursday

Time: 10:00 - 11:30

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

- Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
- Write your seat number in words and figures in the Answer sheet provided to you.
- Do not copy or help anyone in person or by using signs/ digital gadgets while you write the exam/ while you start the exam/ while you end the exam.
- Calculator is allowed.
- Figures to the right indicate maximum marks.
- This question paper comprises of 2 printed sides.
- All questions are compulsory.

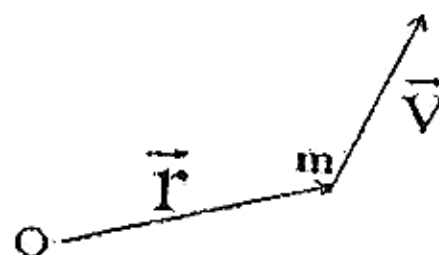
- Q1
- Explain optical pumping in laser? 2Marks CO5 BL2
 - In He-Ne laser which gas transition is responsible for lasing? 2Marks CO5 BL2
 - Calculate the specific charge of an electron? Given that 2Marks CO6 BL2
 $e = 1.6 \times 10^{-19} \text{ C}$
 $m_e = 9.1 \times 10^{-31} \text{ Kg}$
 - How do Fresnel and Fraunhofer diffractions differ from each other? 2Marks CO5 BL2
 - Write the fringe width expression for Young's double slit experiment? 2Marks CO5 BL2
 - In a cathode ray tube electron beam passes through magnetic field 4Marks CO6 BL3
 $B = 0.001 \text{ Tesla}$ and an electric field $E = 1 \times 10^4 \text{ V/m}$ Both electric field and magnetic field acting simultaneously at the same point. If the path of electron beam remains undeviated, calculate the speed of electron.
- Q2
- Define zeroth law of thermodynamics? 2Marks CO4 BL1
 - Find the efficiency of Carnot's engine working between steam point (100°C) and ice point (0°C). 3Marks CO4 BL3
 - In Young's double slit experiment separation of slit is 2 mm and fringe spacing is 0.32 mm at a distance of 1 meter from the slits. Calculate the wavelength of light? 4Marks CO5 BL3

Q3 a) Define simple harmonic motion? A block of mass m is attached with a spring having elastic constant k . The block is pulled and left, to oscillate about its equilibrium position. Write down its equation of motion in differential form and find out its time period?

3Marks CO1 BL1

b) Define angular momentum $\vec{L}$ of a particle of mass ' m ' about the point having velocity $\vec{v}$ as shown in the figure below. Where position vector of the particle from point O is $\vec{r}$. Further find the relation of the torque $\vec{\tau}$ about O ?

4Marks CO2 BL1



c) Define Q value of nuclear reaction?

2Mark CO3 BL1

d) A nuclear reaction is given by ${}^{10}_5\text{B} + {}^4_2\text{He} = {}^{13}_6\text{C} + {}^1_1\text{H}$, where,

4Marks CO3 BL3

$$\left(\begin{array}{l} {}^{10}_5\text{B} = 10.016125 \text{ amu}, {}^4_2\text{He} = 4.003874 \text{ amu} \\ {}^{13}_6\text{C} = 13.007440 \text{ amu}, {}^1_1\text{H} = 1.008146 \text{ amu} \\ \text{and } 1 \text{ amu} = 931.4 \text{ MeV} \end{array} \right)$$

Calculate the Q value of the reaction?

Q4 a) Classify the insulators, conductors, and semiconductors on the basis of band theory and explain it using band diagram?

4Marks CO7 BL2

b) Explain solar Cells?

2Marks CO7 BL2

Q5 Define critical temperature T_c of the superconductor?

3Marks CO7 BL1



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.						
----------	--	--	--	--	--	--

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2022-26

Semester: I

Course: Physics

Course Code: 0705210102

Batch: 2022-26 (Sem I)

Backlog April/Even 2023 Examination

Date: 21/06/2023

Day: Wednesday

Maximum Marks: 45

Time: 10:00 - 11:30

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

- Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
- Write your seat number in words and figures in the Answer sheet provided to you.
- Do not copy or help anyone in person or by using signs/ digital gadgets while you write the exam/ while you start the exam/ while you end the exam.
- Calculator is allowed.
- Figures to the right indicate maximum marks.
- This question paper comprises of 2 printed sides.
- All questions are compulsory.

Q.1. a) Explain the construction working of nuclear reactor. 6 Marks CO1 BL2

Q.1. b) What is isothermal process? Derive an expression for the work done in the isothermal process. 3 Marks CO2 BL2

Q.1. c) Explain the following (Any two) 3 Marks CO2 BL2

- Zeroth law of thermodynamics
- Adiabatic process
- Second law of thermodynamics

Q.2. a) Write the expression for the determination of wavelength of light using Newton's Rings experiment. By using the expression find the wavelength of light if the diameter of 5th ring is 0.3 cm and 25th ring is 0.8 cm. (Radius of curvature of planoconvex lens is 1m) 4 Marks CO3 BL3

- Q.2. b) What is principle of superposition of waves and interference of light? 2 Marks CO3 BL1
- Q.2. c) Explain the various properties of ultrasonic waves. 3 Marks CO3 BL2
- Q.2. d) Calculate the frequency of ultrasonic waves that can be generated by nickel rod of 4 cm length. 2 Marks CO3 BL3
- Given data:
- Young's modulus of nickel = $20.7 \times 10^{10} \text{ N/m}^2$
- Density = 8900 Kg/m^3
- Q.3. a) Draw the V-I characteristics of the P-N junction diode. State the inference obtained from the characteristics and the application of diode based on inference. 4 Marks CO4 BL3
- Q.3. b) Distinguish between conductors, semiconductors and insulators on the basis of band gap theory. 3 Marks CO4 BL2
- (Draw the band diagram for each and show the band gap energy values).
- Q.3. c) Distinguish between stimulated emission and spontaneous emission. 2 Marks CO5 BL2
- Q.3. d) Explain the various characteristics of laser light. 2 Marks CO5 BL2
- Q.4. a) How the nanomaterials are classified on the basis of dimensions? 3 Marks CO5 BL2
- Classify the nanomaterials with suitable examples.
- Q.4. b) Explain the various applications of nanotechnology. 2 Marks CO5 BL2
- Q.4. c) Explain type I and type II semiconductors. 3 Marks CO5 BL2
- Q.4. d) Explain the various applications of superconductivity. 3 Marks CO5 BL2



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.

--	--	--	--	--	--

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2022-26, 2021-25

Semester: II

Course: Programming and Problem Solving

Course Code: 0705210206

Batch: 2022-26 (R) and 2021-25 (B) (Sem II)
Regular and Backlog April/Even 2023
Examination

Date: 17/07/2023

Maximum Marks: 30

Day: Monday

Time: 10:00 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

Q.1 a) Design a flowchart and an algorithm to find greatest of two numbers. 3 CO1, BL3

b) Illustrate concept of computational thinking with real life example. 3 CO1, BL2

Q.2 a) The class has student roll numbers from 1 to 20. 4 CO2, BL3

$G1 = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$G2 = \{5, 6, 7, 8, 9, 10, 11, 12, 13\}$

$G3 = \{1, 2, 6, 7, 11, 12, 15, 16, 20\}$

G1 is a set of roll numbers who play cricket.

G2 is a set of roll numbers who play football.

G3 is a set of roll numbers who play chess.

Write a python program to answer the following questions related to above mentioned sets:

(i) Display the roll number of students who play outdoor games only.

(ii) Display the roll number of students who play all the games.

b) In a multi national company, the HR wants to keep the records of recruited employees. He has to maintain the data as id, and name, age, salary, address. If new employee is recruited he should not get the same emp id. Write a python program to add new employee and retrieve all the data of the existing employees based on his id. (Hint: Use dictionary) 4 CO2, BL3

- Q.3 a) Write a program to find whether a number entered by the user is a spy number or not using user-defined function. (Hint: spy number is a number if the sum and product of its digits are equal e.g 132 is a spy number) 4 CO3, BL3
- b) Write the difference between a user defined function and lambda function with example in python. 4 CO3, BL2
- Q.4 a) Write a python program to create an object of a DOG class with attributes as color, breed and name. Use destructor to release the memory. 4 CO4, BL3
- b) Describe in detail the concept of inheritance and method overriding with examples. 4 CO4, BL3

Batch: 2022-26 (R) and 2021-25 (B) (Sem II)
Regular and Backlog April/Even 2023
Examination



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.

--	--	--	--	--	--

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY(COMPUTER SCIENCE AND ENGINEERING)

Batch: 2021-25,2022-26,2023-27

Semester: II

Course: Programming and Problem Solving

Course Code: 0705210206

Date: 28/05/2024

Maximum Marks: 30

Day: Tuesday

Time: 10:00 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

- a) Use a Ball Pen/ Ink Pen (Blue only) to answer all questions.
- b) Write your seat number in words and figures in the Answer sheet provided to you.
- c) Figures to the right indicate maximum marks. All questions are compulsory.

- Q1. a) Construct a flowchart to determine whether a given number is prime or not. Prompt the user to input an integer, apply the prime number test, and display the result. 3 CO1, BL3
- b) Demonstrate the concept of computational thinking using a real-world scenario. 3 CO1, BL2
- Q2. a) What is a dictionary in Python? Explain any three dictionary methods. 4 CO2, BL2
- b) Write a Python program to remove punctuations from a string entered by the user without using string inbuilt methods. 4 CO2, BL3
- Q3. a) Write a program to find whether a number entered by the user is a spy number or not using user-defined function. (Hint: spy number is a number if the sum and product of its digits are equal e.g 132) 4 CO3, BL3
- b) What is a module in Python? How can you import specific attributes or functions from a module in Python? 4 CO3, BL1
- Q4. a) What is a class in Python? How does it serve as a blueprint for creating objects? 3 CO4, BL1
- b) Write a Python program to model different geometric shapes using inheritance. Define a base class Shape with attributes like length, width, and radius. Then, create subclasses Rectangle and Circle that inherit from the Shape class and display the area of each shape. 5 CO4, BL3



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category -I by UGC

Seat No.						
----------	--	--	--	--	--	--

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY(COMPUTER SCIENCE AND ENGINEERING)

Batch: 2023-27,2024-28

Semester: I

Course: Statistics and Probability

Course Code: 0705210108

Date: 19/12/2024

Day: Thursday

Maximum Marks: 45

Time: 10:00 - 11:30

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

- All questions are compulsory.
- Write your seat number in words and figures in the Answer sheet provided to you.
- Do not copy or help anyone in person or by using signs/ digital gadgets while you write the exam/ while you start the exam/ while you end the exam.
- Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
- Make suitable assumptions wherever necessary.
- Figures to the right indicate full marks.
- A table of area under normal distribution will be provided during exam.
- Non programmable calculator is allowed.

Que.1 Attempt any five parts of the following questions.

- Discuss the inclusive and exclusive form of data representation in tabular form. 3 CO1, BL1
- Write short note on decile, quartile, percentile. 3 CO1, BL1
- Write all the different coefficients of skewness. 3 CO2, BL1
- Define regression, and find the expression for angle between two regression lines. 3 CO4, BL1
- Explain the normal equations for the linear regressions for two variables. 3 CO4, BL2
- Explain Distribution function corresponding to continuous random variable. Write the expression for getting probability $P[1 < X < 5]$ in terms of distribution function. 3 CO6, BL2

g) Discuss the condition under which a Binomial distribution can be approximated as Poisson Distribution. 3 CO6, BL2

h) Define Poisson distribution and write its properties. 3 CO6, BL1

Que.2 a) Calculate Bowley's coefficients of Skewness for the following data 5 CO2, BL4

Observations:	0	1	2	3	4	5
Frequencies :	1	8	28	56	70	56

b) Obtain first four moments (about mean) of the following distribution hence conclude about the Kurtosis 5 CO3, BL3

Observations:	1	2	3	4	5	6
Frequencies :	4	5	20	40	50	65

Que.3 a) Calculate the correlation coefficient using short cut method only between the variables X and Y 5 CO4, BL4

X	10	12	14	18	20
Y	5	6	7	10	12

b) Fit both the straight-line curves (regression lines) with the help of the least-square method. 5 CO5, BL3

X	0	1	2	3	4
Y	1	1.8	3.3	4.5	6.3

Que.4 a) A bag contains 3 red and 5 white marbles. A marble is drawn from the bag and its color is observed. Then the marble is put back into the bag and the contents are thoroughly mixed. Using
(i) Binomial distribution
(ii) Normal approximation
Find the probability that in 20 such drawings a red marble is selected at most 3 times. 5 CO6, BL3

b) A factory produces components of which 1% are defective. The components are packed in boxes of 10. A box is selected at random.
(i) Find the probability that the box contains exactly one defective component. (ii) Find the probability that there are at least 2 defective components in the box. 5 CO6, BL3



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.

--	--	--	--	--	--

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY(COMPUTER SCIENCE AND ENGINEERING)

Batch: 2023-27

Semester: I

Course: Statistics and Probability

Course Code: 0705210108

Date: 29/05/2024

Maximum Marks: 45

Day: Wednesday

Time: 10:00 - 11:30

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

Que.1 Attempt any five parts of the following questions.

- | | | |
|---|---|----------|
| a) Write short note on Histogram, frequency polygon, Ogives. | 3 | CO1, BL3 |
| b) Write short note on primary and secondary data, describe with examples. | 3 | CO1, BL3 |
| c) Describe the different measures of dispersion. | 3 | CO2, BL2 |
| d) Describe the effect of change of origin and change of scale on regression coefficient? | 3 | CO4, BL2 |
| e) Define regression. Write down the properties of regression coefficient. | 3 | CO4, BL1 |
| f) Define continuous random variables and Explain p.d.f with its properties. | 3 | CO6, BL1 |
| g) Define Exponential distribution and its properties. | 3 | CO6, BL1 |
| h) Define (i) Mutually Exclusive events (ii) Exhaustive events. | 3 | CO6, BL1 |

- Que.2 a) Using change of scale and change of origin calculate the coefficient of variation for the following distribution 5 CO2, BL4

Ages (in Years): 20-30 30-40 40-50 50-60 60-70 70-80 80-90

No of members : 5 60 120 150 140 50 5

- b) Obtain first four moments about origin of the following distribution hence find moments about mean 5 CO3, BL3

Observations: 1 3 4 5 7 8 10

Frequencies : 2 6 8 10 14 10 5

- Que.3 a) Ten competitors in a musical test were ranked by the three judges A, B and C in the following order: 5 CO4, BL3

Ranks by A: 1 6 5 10 3 2 4 9 7 8

Ranks by B: 3 5 8 4 7 10 2 1 6 9

Ranks by C: 6 4 9 8 1 2 3 10 5 7

Using rank correlation method discuss which pair of judges has the nearest approach to common liking in music.

- b) The table below is a data set on 8 test scores and hours spent studying for that test. Fit the straight-line curves (regression lines) with the help of the least-square method 5 CO5, BL3

Test Score: 95 90 89 82 80 70 60 95

Hours Spent: 6 4 5 5 4 3 0.5 6

- Que.4 a) A biased dice is tossed 10 times and appearance of odd no or even no is observed with $P(\text{Even no}) = 0.7$ and $P(\text{Odd no}) = 0.3$. Let X be a random variable, which denotes the largest string of, even no appears. Find 5 CO6, BL3

- (i) Probability distribution
- (ii) Mean
- (iii) Variance
- (iv) Coefficient of skewness
- (v) Coefficient of kurtosis

- b) A bag contains two red and six white marbles. A marble is drawn from the bag and its color is observed. Then the marble is put back into the bag and the contents are thoroughly mixed. Using 5 CO6, BL3
- (i) Binomial distribution
 - (ii) Normal approximation
- Find the probability that in 20 such drawings a red marble is selected at most 3 times.



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.

--	--	--	--	--	--	--

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY(COMPUTER SCIENCE AND ENGINEERING)

Batch: 2023-27

Semester: I

Course: Statistics and Probability

Course Code: 0705210108

Date: 27/12/2023

Maximum Marks: 45

Day: Wednesday

Time: 13:30 - 15:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

Que.1 Attempt any five parts of the following questions.

- | | | | |
|----|---|---|----------|
| a) | Write short note on Diagrams and its types?. | 3 | CO1, BL3 |
| b) | Write short note on grouped and ungrouped data, describe with examples. | 3 | CO1, BL3 |
| c) | Describe the different measures of central tendency. | 3 | CO2, BL2 |
| d) | Describe the effect of change of origin and change of scale on Karl person's correlation coefficient? | 3 | CO4, BL2 |
| e) | Define correlation. Write down the properties of correlation coefficient. | 3 | CO4, BL1 |
| f) | Define discrete random variables and Explain p.m.f with its properties. | 3 | CO6, BL1 |
| g) | Define Uniform Distribution and its properties. | 3 | CO6, BL1 |
| h) | Define (i) Equally likely events (ii) Independent event. | 3 | CO6, BL1 |

- Que.2 a) Calculate Bowley's coefficients of Skewness (based on quartiles) for the following data 5 CO2, BL4

Observations:	0	1	2	3	4	5	6	7	8
Frequencies :	1	8	28	56	70	56	28	8	1

- b) Obtain first four moments (about mean) of the following distribution hence conclude about the Kurtosis 5 CO3, BL3

Observations:	1	2	3	4	5	6	7	8	9
Frequencies :	4	5	20	40	50	65	30	10	5

- Que.3 a) Calculate the correlation coefficient using short cut method only between the heights of fathers in inches (X) and their sons (Y) 5 CO4, BL4

X	65	66	57	67	68	69	70	72
Y	67	56	65	68	72	72	69	71

- b) Fit the straight-line curves (regression lines) with the help of the least-square method. 5 CO5, BL3

X	65	66	57	67	68	69	70	72
Y	67	56	65	68	72	72	69	71

- Que.4 a) A biased coin weighted so that $P(H) = 3/7$ and $P(T) = 4/7$, is tossed 20 times. Let X be a random variable, which denotes the largest string of, tells which occurs. Find 5 CO6, BL3

- (i) Probability distribution
- (ii) Mean
- (iii) Variance
- (iv) Coefficient of skewness
- (v) Coefficient of kurtosis

- b) A machine produces bolts, which are 10% defective. Find the probability that in a random sample of 400 bolts produced by the machine 5 CO6, BL3

- (i) Between 30 and 50
 - (ii) At the most 30
 - (iii) 55 or more
- of the bolts will be defective.



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.

--	--	--	--	--	--

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2022-26

Semester: II

Course: Physics

Course Code: 0705210213

Batch: 2022-26 (R) and 2021-25 (B) (Sern II)
Regular and Backlog April/Even 2023
Examination

Date: 12/07/2023

Day: Wednesday

Maximum Marks: 45

Time: 10:00 - 11:30

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

- Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
- Write your seat number in words and figures in the Answer sheet provided to you.
- Do not copy or help anyone in person or by using signs/ digital gadgets while you write the exam/ while you start the exam/ while you end the exam.
- Calculator is allowed.
- Figures to the rights indicate maximum marks.
- This question paper comprise of 2 printed sides.
- All questions are compulsory.

Q1 a) Explain optical pumping in laser?

1Mark CO5 BL2

b) In He-Ne laser which gas transition is responsible for lasing?

1Mark CO5 BL2

c) Calculate the specific charge of an electron? Given that

1Mark CO6 BL2

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$m_e = 9.1 \times 10^{-31} \text{ Kg}$$

d) What is the difference between Fresnel and Fraunhofer diffractions?

1Mark CO5 BL2

e) What is the conditions of Bright fringes (maxima) and Dark fringes (minima) for reflected light from the transparent film having refractive index μ , thickness t and angle of refraction ϕ .

2Marks CO5 BL2

f) In a cathode ray tube electron beam passes through magnetic field $B = 0.005 \text{ Tesla}$ and an electric field $E = 5 \times 10^4 \text{ V/m}$ Both electric field and magnetic field acting simultaneously at the same point. If the path of electron beam remains undeviated, calculate the speed of electron.

3Marks CO6 BL3

Q2 a) Define zeroth law of thermodynamics?

2Marks CO4 BL1

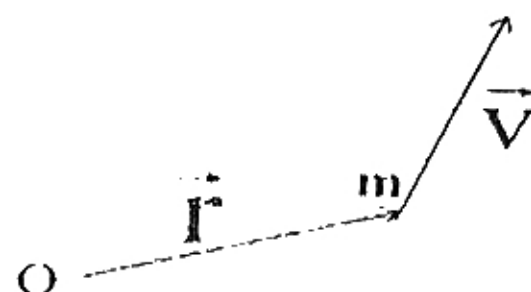
b) Find the efficiency of Carnot's engine working between steam point (100°C) and ice point (0°C)?

2Marks CO4 BL3

- c) In Young's double slit experiment separation of slit is 2 mm and fringe spacing is 0.32 mm at a distance of 1 meter from the slits. Calculate the wavelength of light? 2Marks CO5 BL3
- d) Define Rayleigh criterion for resolution? Calculate the aperture of the objective of the telescope, which is used to resolve stars separated by 4.88×10^{-6} radian for light wavelength 6×10^{-7} meter? 3Marks CO5 BL3

- Q3 a) Define simple harmonic motion? A block of mass m is attached with a spring having elastic constant k . The block is pulled and left, to oscillate about its equilibrium position. Write down its equation of motion in differential form and find out its time period? 3Marks CO1 BL1

- b) Define angular momentum $\vec{L}$ of a particle of mass ' m ' about the point having velocity $\vec{v}$ as shown in the figure below. Where position vector of the particle from point O is $\vec{r}$. Further find the relation of the torque $\vec{\tau}$ about O ? 3Marks CO2 BL1



- c) Define Q value of nuclear reaction?

- d) A nuclear reaction is given by ${}^{10}_5\text{B} + {}^4_2\text{He} = {}^{13}_6\text{C} + {}^1_1\text{H}$, where,

$$\left(\begin{array}{l} {}^{10}_5\text{B} = 10.016125 \text{ amu}, {}^4_2\text{He} = 4.003874 \text{ amu} \\ {}^{13}_6\text{C} = 13.007440 \text{ amu}, {}^1_1\text{H} = 1.008146 \text{ amu} \\ \text{and } 1 \text{ amu} = 931.4 \text{ MeV} \end{array} \right)$$

Calculate the Q value of the reaction?

1Mark CO3 BL1
2Marks CO3 BL3

- Q4 a) Classify the insulators, conductors, and semiconductors on the basis of band theory and explain it using band diagram? 3Marks CO7 BL2

- b) What are solar Cells?

- c) What are nanomaterials?

- d) What are nanomaterials properties which is different from bulk materials?

1Mark CO7 BL2
1Mark CO7 BL2
2Marks CO7 BL2

- e) What are liquid crystals?

2Marks CO7 BL2

- Q5 a) What is magnetostriction effect?

- b) Define critical temperature T_c of the superconductor?

- c) Define type I and type II superconductors?

- d) Define isothermal and adiabatic processes?

2Marks CO7 BL2
3Marks CO7 BL1
2Marks CO7 BL1
2Marks CO7 BL1

Batch: 2022-26 (R) and 2021-25 (B) (Sem II)
Regular and Backlog April/Even 2023
Examination



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.

--	--	--	--	--	--

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2021-25

Semester: III

Course: Fundamentals of Data Structures

Course Code: 0705210303

Batch: 2021-25 (B) (Sem III)
Backlog April/Even 2023 Examination

Date: 03/07/2023

Maximum Marks: 45

Day: Monday

Time: 10:00 - 11:30

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

Q.1 a) Declare a 3D SIT_GRID (Integer type) that has the dimension 128x256x256. Write a C program to assign the memory to SIT_GRID dynamically. Initialize all the elements of SIT_GRID to 255 for first 64 slices and assign 0 to the next 64 slices respectively. 5 CO1, BL3

Q.1 b) How many times is the if statement in the inner for loop executed in the following code snippet? Give your answer as a function of n with proper explanation. 5 CO1, BL2

```
for (int i = 0; i < n - 1; i++)
{
    int min = i;
    for (int j = i + 1; j < n; j++)
    {
        if (a[j] < a[min])
        {
            min = j;
        }
    }
    int temp = a[i];
    a[i] = a[min];
    a[min] = temp;
}
```

Q.2 a) Given two sorted lists, L_1 and L_2 , in ascending order. Give an efficient algorithm to find the intersection of the two lists. The Intersection of L_1 and L_2 is the set of elements that is present in both the list. 7 CO2, BL4

Q.2 b) Perform Radix sort on the array $A[] = \{23, 14, 6, 92, 63, 27, 79, 26, 90, 5\}$ and write down the output of each pass of the Radix sort. What is the worst case complexity of the Radix sort? 3 CO2, BL3

Q.3 a) Write a C program or design the algorithms to implement the `is_Empty_MS(int st_number)`, `is_Full_MS(int st_number)`, and `Push_MS(int x, int st_number)` for multiple stacks (MS) using an array. Here, `st_number` denotes the stack number. 5 CO4, BL3

Q.3 b) Consider the following recursive function definition.

```
int SIT_2(int n)
{
    if(n>0)
    {
        print(n);
        SIT_2(n-1);
        print(n-1);
        SIT_2(n-2);
        print(n-2);
    }
}
```

What is the output of the function call SIT_2(3)? How many function calls are required to evaluate SIT_2(4)? 5 CO4, BL3

Q.4 a) What is the output of the following code snippet? Justify your answer by showing the stack and queue entries. 4 CO3, BL3

```
char S[] = "RUPGANTIS_ta_SDF"
len = strlen(S); // length of the string S
for(int i = 0; i<len; i++)
{
    SIT_S.push(S[i]);
}
while(!SIT_S.isEmpty())
{
    SIT_Q.enqueue(SIT_S.pop());
}
while(!SIT_Q.isEmpty())
{
    printf("%c", SIT_Q.dequeue());
}
```

Q.4 b) Consider the following sequence of operations on an empty stack: push(54); push(52); pop(); push(55); push(62); s = pop(); Consider the following sequence of operations on an empty queue: insert(21); insert(24); delete(); insert(28); insert(32); q = delete(); Draw the final state of the stack and queue and find the value of s+q. 3 CO3, BL4

Q.4 c) A letter means to push and an asterisk means pop in the following sequence. Give the sequence of values returned by the pop operations when this sequence of operations is performed on an initially empty stack. E A S * Y * Q U E * * * S T * * * I O * N * * * 3 CO3, BL3

Q.5 Write a C program demonstrating the various file operations in C. Improve your C program with different file opening modes. 5 CO5, BL2



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.

--	--	--	--	--	--

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2021-25,2022-26

Semester: III

Course: Fundamentals of Data Structures

Course Code: 0705210303

Date: 07/12/2024

Maximum Marks: 45

Day: Saturday

Time: 13:30 - 15:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

- a) Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
- b) Write your seat number in words and figures in the Answer sheet provided to you.
- c) Do not copy or help anyone in person or by using signs/ digital gadgets while you write the exam/ while you start the exam/ while you end the exam.
- d) Calculator is allowed.
- e) Figures to the rights indicate maximum marks.
- f) This question paper comprises of 1 printed side.
- g) All questions are compulsory.
- h) Marks for each questions are mentioned next to it.

- Q1 a) Differentiate between bubble sort and insertion sort 5 CO1 BL4
b) Explain how the choice of pivot affects the efficiency of the Quick Sort algorithm. 5 CO1 BL2
- Q2 a) Mention advantages and practical applications of using a circular linked list over other types of linked lists 5 CO2 BL3
b) Brief different operations of queue. 5 CO2 BL2
- Q3 a) Explain ADT in short 5 CO3 BL2
b) Given the preorder traversal G B Q A C K F and the inorder traversal Q B G K C A F of a binary tree, construct the binary tree and determine the postorder traversal. Explain each step of the process 5 CO3 BL3
- Q4 a) Differentiate between BFS and DFS 5 CO4 BL4
b) Explain the concept of a Minimum Spanning Tree 5 CO4 BL2
- Q5 What is AVL trees? Explain its key properties. 5 CO5 BL2



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under Section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A' Grade (3.58/4) Awarded Category - I by UGC

Seat No.					
----------	--	--	--	--	--

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2021-25

Batch: 2021-25 (Sem III)

Semester: III

Regular October/Odd 2022 Examination

Course: Fundamentals of Data Structures

Course Code: 0705210303

Date: 18/01/2023

Maximum Marks: 45

Day: Wednesday

Time: 10:00 - 11:30

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

Q.1 a) What is meant by "fast algorithm"? Justify your answer with appropriate examples. 3 CO1, BL2

Q.1 b) Declare a struct STUDENT containing a student's Name, Age, and Address (Use appropriate datatypes). Write a C program to create an array of structures STUDENT for ten students. Your program should accept input for all ten students and print their information to the console. 5 CO1, BL3

Q.1 c) Declare a union FACULTY containing a faculties's Name, Age, and Subject_name (Use appropriate datatypes). Write a declaration to create an array of unions FACULTY for ten faculties. What is the size of union FACULTY and the array of unions FACULTY for ten faculties. 2 CO1, BL3

Q.2 a) Consider two arrays, $A[n]$ and $B[m]$, where n is significantly greater than m . An array A is an unsorted array, whereas an array B is a sorted array. Design an efficient algorithm or write a C program that finds the common values in both arrays. Comment on the time complexity of your algorithm or C program. 7 CO2, BL4

Q.2 b) Perform Radix sort on the array $A[] = \{328, 96, 435, 934, 772, 891, 690, 563, 5, 257, 149, 18\}$ and write down the output of each pass of the Radix sort. Why is the Radix sort not preferred over other sorting techniques? 3 CO2, BL3

Q.3 Consider the following recursive function definition.

```
int SIT_1(int n)
{
    if(n==0) return 0;
    else if(n==1) return 1;
    else return (SIT_1(n-1)+SIT_1(n-2));
}
```

- Q.3 a) What output you will get for the function calls SIT_1(8), and SIT_1(10) respectively? 3 CO4, BL4
- Q.3 b) How many function calls are required to evaluate the function calls SIT_1(8), and SIT_1(9) respectively? 3 CO4, BL4
- Q.3 c) How many additions are required to evaluate the function calls SIT_1(9), and SIT_1(10) respectively? 3 CO4, BL4
- Q.3 d) How many function calls are required to evaluate the function calls SIT_1(SIT_1(6))? 1 CO4, BL4
- Q.4 a) What is the output of the following code segment? 5 CO3, BL4

```
int item1 = 100;
int item2 = 200;
int item3 = 300;
SIT_Q_Enque(item2);
SIT_Q_Enque(item1);
SIT_Q_Enque(item1 + item3);
item2 = SIT_Q_Deque();
SIT_Q_Deque();
SIT_Q_Enque(item3 * item3);
SIT_Q_Enque(item2);
SIT_Q_Enque(3);
item1 = SIT_Q_Deque();
SIT_Q_Deque();
printf("%d", item1 + item2 + item3);
while (! SIT_Q.empty())
{
    printf("%d", SIT_Q_Deque());
}
```

- Q.4 b) Write a C program or design the algorithms to implement the is_Empty_C(), deque_C(), and enqueue_C(int x) for circular queue using an array. 5 CO3, BL2
- Q.5 a) A file named "input.txt" contains only positive integers. Write a program to count the number of even and odd numbers in the file input.txt on the console. 4 CO5, BL3
- Q.5 b) Enlist the file opening modes in C. 1 CO5, BL1



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.

--	--	--	--	--	--

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2022-26

Semester: II

Course: Physics

Course Code: 0705210213

Date: 22/05/2024

Maximum Marks: 45

Day: Wednesday

Time: 10:00 - 11:30

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

- Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
- Write your seat number in words and figures in the Answer sheet provided to you.
- Do not copy or help anyone in person or by using signs/ digital gadgets while you write the exam/ while you start the exam/ while you end the exam.
- Calculator is allowed.
- Figures to the right indicate maximum marks.
- This question paper comprises of 2 printed sides.
- All questions are compulsory.

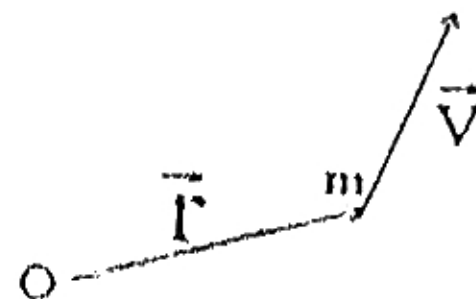
- Q1
- Explain optical pumping inversion? 1Mark CO5 BL2
 - Write any two medical applications of laser? 1Mark CO5 BL1
 - A beam of positive ions is deflected by a magnetic field, what would be effect on their kinetic energy? 1Mark CO6 BL2
 - Define Interference of light? 1Mark CO5 BL1
 - Newton's rings observed with wavelength 5.0×10^{-7} meter and diameter of the 10th dark ring is 0.6 cm. find the radius of curvature of the lens? 2Marks CO5 BL3
 - An electron having velocity 10^6 m/sec experiences maximum force of 3.2×10^{-14} Newton, when it enters in a uniform magnetic field perpendicular to its direction. What is the value of magnetic field? Where charge of the electron $e = 1.6 \times 10^{-19}$ C. 3Marks CO6 BL3
- Q2
- Define Zeroth law of thermodynamics? 3Marks CO4 BL1
 - A parallel beam of light incident on a plane transmission grating having 5000 lines/cm and second order spectral line is observed to be deviated by 60° calculate the wavelength of spectral lines? 3Marks CO5 BL3

- c) Define Piezo electric effect? Calculate the natural frequency of ultrasonic waves using the following data: Thickness of quartz plate $t = 1.5 \times 10^{-3}$ m, Young's modulus of quartz is $Y = 8.0 \times 10^{10}$ N/m² and density of quartz is $\rho = 2.65 \times 10^3$ kg/m³ 3Marks CO3 BL2

- Q3 a) Derive the time period of the block of mass m attached with the spring having spring constant k . If this block is pulled one side and released for oscillations. 3Marks CO1 BL3

- b) In above question No (3 a), motion of the block of mass m , will be periodic or simple harmonic and why? 2Marks CO2 BL2

- c) Calculate the angular momentum $\vec{L}$ and torque $\vec{\tau}$ of a particle of mass $m = 10$ kg about the point O having velocity $\vec{v} = 2\hat{i} + 4\hat{j}$ as shown in the figure below. Where position vector of the particle from point O is $\vec{r} = 3\hat{i} + 5\hat{j}$. 4Marks CO2 BL3



- Q4 a) Define Q value of nuclear reactor? 3Marks CO3 BL1
 b) What is nuclear reactor? Name the components of nuclear reactor? 3Marks CO3 BL1
 c) What are the physical properties of nanomaterials? 3Marks CO7 BL1

- Q5 a) Explain Meissner Effect? 3Marks CO7 BL2
 b) Define cooper pairs? 3Marks CO7 BL1
 c) Explain London penetration depth? 3Marks CO7 BL2
